NI TB GENOMIC SURVEILLANCE

Proof of Concept System

Step-by-Step User Guide

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This guide provides current instructions for deploying, configuring, and operating the TB Genomic Surveillance PoC system on Windows using the updated workflow.

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1. SYSTEM REQUIREMENTS

Operating System: Windows 10 or later

Python: Python 3.10 or later

PostgreSQL: PostgreSQL 12 or later

RAM: Minimum 4 GB (8 GB recommended)

Disk Space: 500 MB for application + 1 GB for data

Network: Localhost access only (development mode)

2. INSTALLATION & SETUP

Step 2.1: Clone or Extract Project

Extract the TB-Genomics project folder to your desktop.

```
C:\Users\YourUsername\Desktop\TB-Genomics-main
```

Step 2.2: Create Python Virtual Environment

Open PowerShell in the project directory and create a Python virtual environment:

```
python -m venv .venv
```

Step 2.3: Activate Virtual Environment

Activate the virtual environment in PowerShell:

```
.\.venv\Scripts\Activate.ps1
```

You should see (.venv) at the start of the command line.

Step 2.4: Install Python Dependencies

With the virtual environment active, install required packages:

```
pip install -r backend/requirements.txt
```

This installs FastAPI, SQLAlchemy, matplotlib, networkx, scipy, reportlab, and other dependencies.

3. DATABASE CONFIGURATION

Step 3.1: Start PostgreSQL Service

Ensure PostgreSQL is running. Open Services (services.msc) and verify that 'postgresql-x64' service is started.

Step 3.2: Connect to PostgreSQL

Open PowerShell and connect to PostgreSQL. If psql is not on PATH, use the full executable path:

```
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U postgres -h localhost -d
postgres
```

Enter your PostgreSQL password when prompted.

Step 3.3: Create Database User & Database

In psql, run these commands:

```
CREATE USER tb WITH PASSWORD 'tb';
CREATE DATABASE tb_surveillance OWNER tb;
GRANT ALL PRIVILEGES ON DATABASE tb_surveillance TO tb;
```

Step 3.4: Load Database Schema

Exit psql (type \q) and load the schema. In PowerShell use the -f flag instead of shell redirection:

```
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U tb -h localhost -d
tb_surveillance -f .\db\schema.sql
```

When prompted for password, enter 'tb'.

If psql is already on PATH, you can remove the full executable path and keep the same arguments.

4. BACKEND SERVER CONFIGURATION

Step 4.1: Start Backend Server

In PowerShell (with virtual environment active), start the FastAPI backend server:

```
uvicorn backend.app:app --host 0.0.0.0 --port 8000 --reload
```

You should see output indicating the server is running on http://0.0.0.0:8000

Step 4.2: Verify Backend Health

Open your browser and visit:

```
http://localhost:8000/docs
```

You should see the FastAPI interactive API documentation page.

Note: Leave this terminal running. Open a new PowerShell window for the next steps.

5. FRONTEND SERVER SETUP

Step 5.1: Navigate to GUI Directory

In a new PowerShell window, navigate to the GUI directory:

```
cd C:\Users\YourUsername\Desktop\TB-Genomics-main\gui
```

Step 5.2: Start Frontend Server

Start the Python HTTP server on port 8081:

```
python -m http.server 8081
```

Step 5.3: Access the Application

Open your browser and visit:

```
http://localhost:8081
```

You should see the NI TB Genomic Surveillance interface with workflow steps for ingest, analysis, search, and governance.

6. GENERATING SYNTHETIC DATA

Step 6.1: Using the Web Interface

In the browser on http://localhost:8081:

1. Navigate to **Step 2 — Upload data**
2. Set **Cases: 250** (or desired number)
3. Set **Seed: 42** (for reproducible results)
4. Check **Reset existing synthetic data first**
5. Click **Generate synthetic dataset**
6. Wait for confirmation message

What Happens:

- The system fetches current TB incidence data from the World Bank API
- 250 realistic synthetic TB cases are generated using country-weighted distribution
- Each case is assigned lineage, resistance profile, and location
- Cases are organized into 4-6 outbreak clusters
- All data is pseudonymised and stored in the database

7. RUNNING ANALYSIS JOBS

Step 7.1: Run Clustering Analysis

In **Step 3 — Run analysis**:

1. Click **Run clustering**
2. A progress bar will appear and update as the job runs
3. When complete, status shows **Completed**

Result: Cases are analyzed for genetic similarity and grouped into clusters.

Step 7.2: Generate Outbreaker2 Inputs

In **Step 3 — Run analysis**:

1. Click **Generate outbreaker2 inputs**
2. Wait for completion

Result: Case data and DNA sequences are prepared for outbreak investigation analysis.

Step 7.3: Run Outbreaker2 Analysis

In **Step 3 — Run analysis**:

1. Click **Run outbreaker2**
2. This may take 30-60 seconds
3. Status will show when analysis completes

Result: Outbreak analysis completes and generates diagnostic plots, an enhanced transmission network, and a PDF outbreak report.

Note: If R/outbreaker2 is unavailable, the system falls back to a mock report generator so the workflow remains usable for demos.

8. INTERPRETING RESULTS

Step 8.1: View Case Summary

In **Step 4 — Results**:

1. Click **View cases**

2. A raw JSON view of all cases is displayed

Include fields: case ID, specimen date, region, lineage, resistance profile

Step 8.2: View Outbreak Analysis Plots

In **Step 4 — Results**:

1. Click **View outbreaker2 summary**

2. The results panel displays:

• **Case Summary:** Total cases, clustered cases, unclustered cases
• **Outbreak Analysis:** MCMC analysis status and likelihood statistics
• **Transmission Network Insights:** Node count, high-confidence links, and ranked priority spreaders
• **Diagnostic Plots:**
  - **MCMC Trace:** Shows convergence of likelihood estimate
  - **Posterior Distributions:** 4-panel histogram of key parameters
  - **Transmission Tree:** Network diagram showing inferred transmission chains
  - **Phylogenetic Tree:** Genetic similarity clustering of isolates
  - **Resistance Heatmap:** Drug resistance pattern overview across cases

Step 8.3: Download the Outbreak Report PDF

In **Step 4 — Results**:

1. Click **Download outbreak report (PDF)**

2. Open the generated report

3. The updated report preserves image aspect ratios so plots are not stretched or squashed

Step 8.4: View Governance & Audit Trail

In **Step 5 — Governance & Compliance**:

1. Click **View audit trail**

2. A detailed log of all system actions is displayed with timestamps

This provides accountability for all data processing and analysis operations.

9. TROUBLESHOOTING

■ ***Browser cannot connect to backend***

✓ Ensure backend is running (uvicorn command). Check firewall settings. Verify port 8000 is available.

■ ***Database connection error***

✓ Verify PostgreSQL is running. Check user 'tb' exists with password 'tb'. Ensure database 'tb_surveillance' was created.

■ ***psql command is not recognized***

✓ Use the full PostgreSQL executable path or add the PostgreSQL bin directory to PATH. In PowerShell, load schema with `psql -f .\db\schema.sql`, not `< db/schema.sql`.

■ ***Synthetic data generation fails***

✓ Check internet connection (World Bank API is called). If offline, system uses fallback values. Verify database is accessible.

■ ***Analysis jobs timeout or fail***

✓ Check disk space in exports/ folder. Verify all Python dependencies are installed. Try regenerating synthetic data.

■ ***Graphics not displaying***

✓ Ensure matplotlib is installed. Check exports/ folder contains PNG files. Try refreshing browser (Ctrl+F5).

■ ***PDF images look distorted***

✓ Regenerate the outbreak report from the updated backend. The current report code scales images proportionally to preserve aspect ratio.

NEXT STEPS & SUPPORT

After PoC Validation:

- Deploy to test environment
- Connect to real TB genomic data sources
- Configure authentication (LDAP/AD integration)
- Set up database backup and recovery procedures
- Deploy Outbreaker2 R package on analysis servers
- Implement production monitoring and logging

Support Resources:

- FastAPI Documentation: https://fastapi.tiangolo.com
- Outbreaker2 Package: https://www.repidemicsconsortium.org/outbreaker2
- PostgreSQL Documentation: https://www.postgresql.org/docs